Exploratory Data Analysis

```
library(tidyverse)
library(knitr)
```

1. Read the data

```
final_data <- read.csv("../data/processed/final_data.csv")
eda_data <- final_data |>
    filter(year>= 2000, year<=2019)
```

2. Overview of the dataset

```
dim(eda_data)
```

```
[1] 3732   17
```

```
names(eda_data)
```

```
 [1] "iso"                "year"               "armed_conflict"
 [4] "maternal_mortality" "infant_mortality"   "under5_mortality"
 [7] "neonatal_mortality" "drought"            "earthquake"
[10] "gdp_1000"           "oecd"               "pop_dens"
[13] "urban"              "age_dep"            "male_edu"
[16] "temp"               "rainfall_1000"
```

```
head(eda_data)
```

```
  iso year armed_conflict maternal_mortality infant_mortality under5_mortality
1 AFG 2000              1               1450             90.5            129.2
2 AFG 2001              1               1390             87.9            125.2
3 AFG 2002              1               1300             85.3            121.1
4 AFG 2003              1               1240             82.7            116.9
5 AFG 2004              1               1180             80.0            112.6
6 AFG 2005              1               1140             77.3            108.4
  neonatal_mortality drought earthquake  gdp_1000 oecd pop_dens    urban
1               60.9       1          0        NA    0 14.13654 16.25324
2               59.7       0          1        NA    0 14.23156 16.25661
3               58.5       0          1 0.1835328    0 14.32270 16.42654
4               57.2       0          1 0.2004626    0 14.40691 16.60701
5               55.9       0          1 0.2216576    0 15.21947 16.71367
6               54.6       0          1 0.2550551    0 15.33619 16.85096
   age_dep male_edu     temp rainfall_1000
1 108.3466 2.762086 12.69959     0.2763704
2 108.9899 2.856936 12.85570     0.2793079
3 109.3472 2.954241 12.71081     0.3805710
4 109.4475 3.054121 12.16592     0.4288939
5 109.2868 3.156706 13.04643     0.3754336
6 107.9646 3.262133 12.23141     0.4415680
```

3. Look at missing data

```
colSums(is.na(eda_data))
```

```
              iso              year     armed_conflict maternal_mortality
                0                 0               2708               438
  infant_mortality   under5_mortality neonatal_mortality           drought
                32                32                 32              3132
        earthquake          gdp_1000              oecd          pop_dens
              3132                74                 12                32
             urban           age_dep          male_edu              temp
                32                12                 32                32
     rainfall_1000
                32
```

Since my EDA focuses on maternal mortality and armed conflict, I also examined missing values specifically for those two variables.

```
eda_data |>
     summarise(
    missing_maternal_mortality = sum(is.na(maternal_mortality)),
    missing_armed_conflict = sum(is.na(armed_conflict))
)
```

```
  missing_maternal_mortality missing_armed_conflict
1                        438                   2708
```

4. Distribution of maternal mortality

First, I examined summery statistics for maternal mortality.

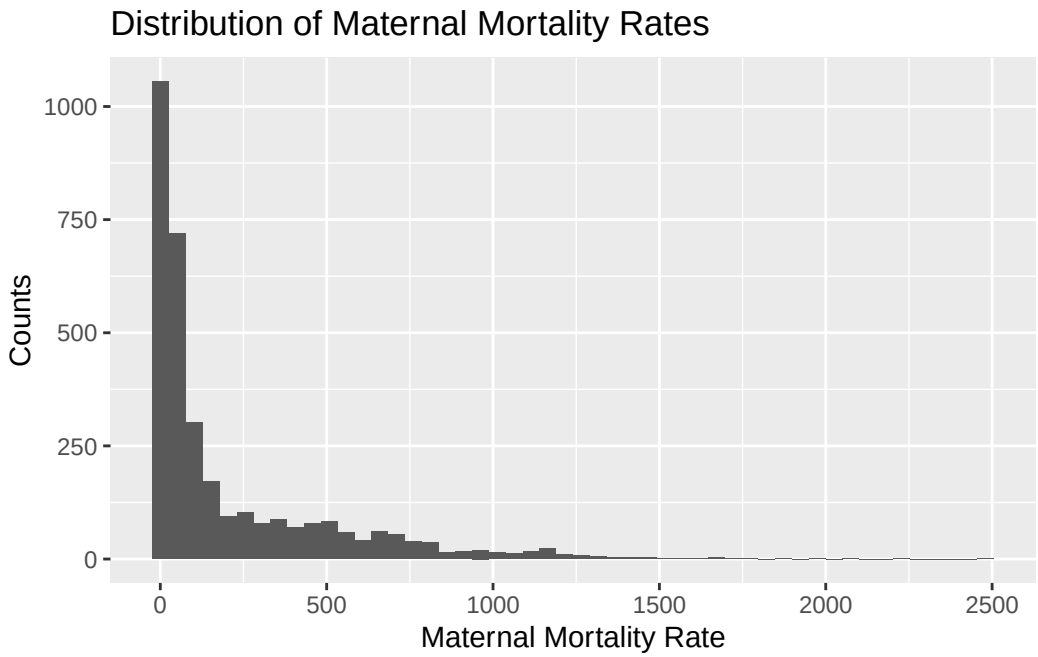
```
summary(eda_data$maternal_mortality)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
    2.0    17.0    66.0   210.6   299.8  2480.0     438
```

Then I Looked at the distribution using a Histogram plot.

```
eda_data |>
    ggplot(aes(x=maternal_mortality)) +
    geom_histogram(bins = 50) +
    labs(
        title = "Distribution of Maternal Mortality Rates",
        x = "Maternal Mortality Rate",
        y = "Counts"
    )
```

```
Warning: Removed 438 rows containing non-finite outside the scale range
(`stat_bin()`).
```



5. Maternal Mortality and Armed Conflict

Next, I compared maternal mortality rates for country-year observations with and without armed conflict. Here, an armed conflict value of 0 represents no armed conflict and a value of 1 represents the presence of armed conflict.

```
eda_data |>
    filter(!is.na(armed_conflict))|>
    ggplot(
        aes(
            x = factor(armed_conflict),
            y = maternal_mortality
        )
    )+
        geom_boxplot()+
        labs(
            title = "Maternal Mortality by Armed Conflict Status",
            x = "Armed Conflict",
            y = "Maternal Mortality Rates"
        )
```

```
Warning: Removed 111 rows containing non-finite outside the scale range
```

(`stat_boxplot()`).

6. Summary by Armed Conflict Status

I calculated the mean and median maternal mortality rates for observations with and without armed conflict.

```
conflict_summary <- eda_data|>
    filter(
        !is.na(armed_conflict),
        !is.na(maternal_mortality)
    )|>
        group_by(armed_conflict)|>
        summarise(
            N = n(),
            Mean_Maternal_Mortality = mean(maternal_mortality),
            Median_Maternal_Mortality = median(maternal_mortality)
        )

    kable(
        conflict_summary,
        caption = "Maternal Mortality by Armed Conflict",
```

```
        digits = 1
    )
```

Table 1: Maternal Mortality by Armed Conflict

armed_conflict	N	Mean_Maternal_Mortality	Median_Maternal_Mortality
0	458	263.0	119
1	455	468.4	354

7. Findings

Maternal mortality rates showed a strongly right-skewed distribution. Among the available observations, the median maternal mortality rate was 66, while the mean was substantially higher at 210.6. Values ranged from 2 to 2480, suggesting that a relatively small number of country-year observations had very high maternal mortality rates. There were 438 missing observations for maternal mortality.

There was also substantial missingness in the armed conflict variable, with 2708 missing observations. Therefore, the comparison between armed conflict groups was based only on observations with non-missing values for both armed conflict and maternal mortality.

Among these complete observations, maternal mortality was higher in country-year observations classified as experiencing armed conflict. The mean maternal mortality rate was 468.4 for observations with armed conflict compared with 263 for observations without armed conflict. Similarly, the median was 354 for the armed conflict group compared to 119 for the non-conflict group.

The boxplot showed the same general pattern, with the distribution of maternal mortality shifted upward among observations with armed conflict. The armed conflict group also appeared to have greater variability and included several observations with particularly high maternal mortality rates.

Overall, this exploratory analysis suggests that higher maternal mortality rates are associated with the presence of armed conflict in the data set. However, these results are descriptive and do not establish a causal relationship. The substantial amount of missing data, particularly for the armed conflict variable, should also be considered when interpreting these findings.